A design overview of the playable prototype

MATH KINGDOM

The Chaos of Numbers

A fantasy RPG in which arithmetic is not a quiz between fights. It is the fight.

Ten heroes · nine regions · thirteen legendary artifacts · one warlock who refuses to lie to you

THE WARD

105

9 6 7

$9 + 6 \times 7 \neq 105$

WHY THIS GAME EXISTS

Most maths games hide the maths behind the fun

× The usual pattern

- Kill an orc. Answer 7 × 8. Kill another orc.
- The question is a tollbooth: pay it, then resume playing.
- Delete the arithmetic and the game still works — which means it was never really there.
- Children learn to endure the maths to reach the reward.

✓ What Math Kingdom does

- Each foe raises a Ward: one number.
- You are dealt tiles and build an expression that equals it.
- Which tiles to spend and which operators to chain IS the tactical decision.
- Delete the arithmetic and there is no combat system left at all.

CORE LOOP

One turn, five beats

Roughly twenty seconds. Repeated a few thousand times across a playthrough.

1 → 2 → 3 → 4 → 5

The deal

Five to eight number tiles land in your hand.

The Ward

The foe raises a target number, generated from your own hand so a route always exists.

The build

Chain tiles and operators. A running total updates as you go.

The cast

Exact match breaks the Ward whole. Close grazes. Wild scatters.

The answer

The foe strikes back. New hand, new Ward, harder or easier than the last.

WORKED EXAMPLE

Anatomy of a single cast

YOUR HAND

9	6	7	4	11

YOU BUILD

$$9 + 6 \times 7$$

The spell resolves the way arithmetic actually resolves — multiplication and division first, left to right, then addition and subtraction.

51

$6 \times 7 = 42$

$9 + 42 = 51$

Not 105.

A player who reads left to right expects 105. The game does not, and neither does mathematics. That gap is the single most useful teaching moment in the design — and the warlock names it out loud when you miss.

FAILURE DESIGN

Three outcomes, none of them a dead end

Being wrong costs you a turn. It never costs you the run, and it always tells you something.

=

Exact

`Full damage`

Perfect resonance. The Ward breaks whole and the relic effects that key off exactness all fire.

≈

Within 2

`40% damage`

A graze. You were close, and the game says by how much and in which direction.

≠

Anything else

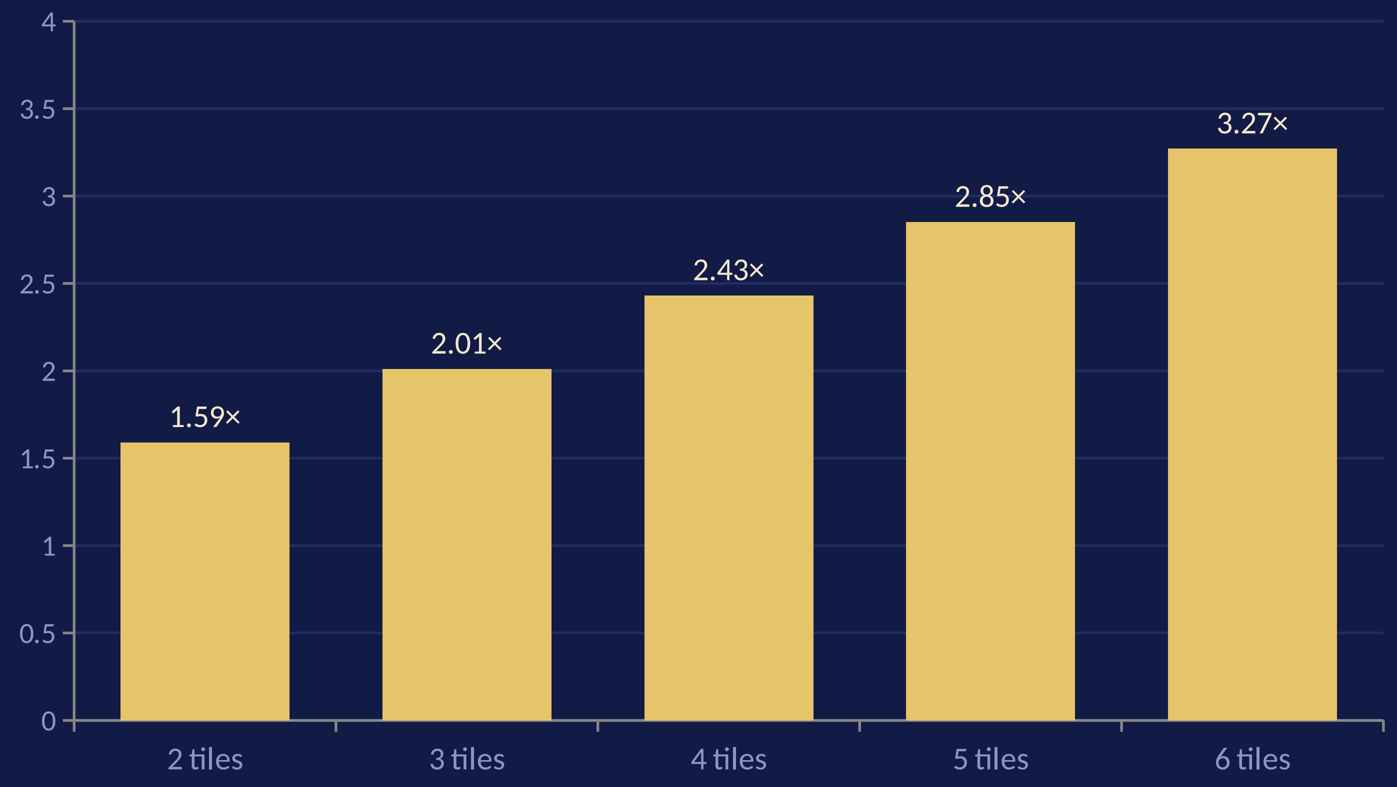
`15% damage`

Scattered. Still a hit — the floor is never zero, because a zero teaches nothing.

INCENTIVE STRUCTURE

Longer spells hit harder

Damage scales with the number of tiles bound into one expression, so the game rewards ambition over safety.



Why this matters

- A two-tile answer is always available and always weakest.
- Reaching a Ward four ways and picking the longest is a strategy problem built out of arithmetic.
- Classes and relics stack on top: the Sigma Sentinel gains a further 18% per tile past the second.

Five ways of being right

Every class biases you toward a different operator, so the same Ward is a different puzzle depending on who you are.

Number Knight

Hits harder on exact matches. Double Strike doubles the next spell.

Equation Wizard

An extra tile every turn. Rewrite deals a fresh hand and Ward.

Geometry Guardian

Highest health, takes less damage. Bulwark absorbs 40.

Fraction Ranger

+25% on any spell that divides. Divide and Conquer cuts a quarter off the foe.

Logic Rogue

+30% on spells of four or more tiles. Deduce writes a working solution.

Five more, added by request

The later five push further into number theory, estimation and precedence.

Prime Berserker

+55% when every tile in the spell is prime. Indivisible turns lost health into damage.

Sigma Sentinel

+18% per tile past the second. Total Sum shields for twice the hand left over.

Modulo Monk

Ten classes means ten different optimal reads of the same hand — which is the point. There is no single correct way to reach a number.

Two extra Study uses and stronger healing. Empty Hand redeals tiles, keeps the Ward.

Decimal Duelist

Near misses graze at 80%. Round Up forces one near miss to count as exact.

Bracket Warden

Brackets before the Tower is built, +35% when using them. Reorder redraws the hand.

THE WORLD

Nine regions, three foes each

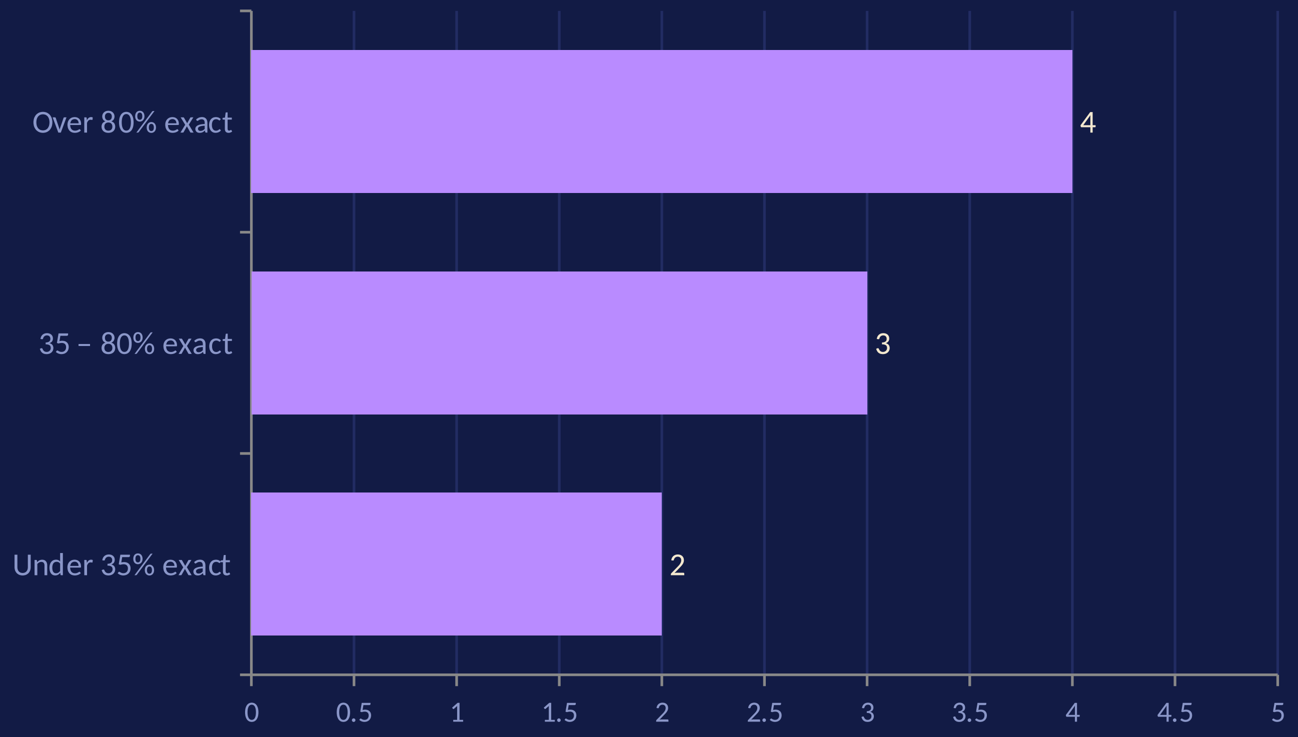
Regions unlock in sequence. Each one narrows the operator set and the number range around a single idea.

+ **Addition Forest** Number bonds	− **Subtraction Swamp** Negative numbers	× **Multiplication Mountains** Times tables, factors
÷ **Division Desert** Factors, remainders	½ **Fraction Kingdom** Parts of a whole	∠ **Geometry City** Area, perimeter, angles
x **Algebra Fortress** Solving for the unknown	% **Probability Islands** Percentages, ratio, chance	0 **Final Number Realm** Everything, at once

DIFFICULTY

The game watches you and moves

Ward complexity is not a menu setting. It tracks exact-match accuracy inside the current fight.



Inside a battle

Three casts at over 80% and the Ward grows a term. Below 35% and it simplifies. The log says so out loud, without scolding.

Between battles

A separate global dial shifts the starting complexity of every future fight up or down one step.

Never below floor

Wards stay at two terms minimum. Struggling makes the game gentler, not emptier.

ENEMIES

Foes with mathematical weaknesses, not just health bars

÷

Weak to division

Sand Divider, Divide Djinn, Half Hound, Odds Corsair. Spells that divide deal 35% more.

≥3

Weak to complexity

Every boss. Spells of three or more tiles deal 30% more, so the finale of each region demands longer thinking.

–

Plain foes

No weakness. These are where a player practises the region's core operator without a tactical wrinkle.

The nine bosses

- Glitch Goblin
- Negative Number Knight
- Factor Titan
- Divide Djinn
- Fraction Phantom
- Architect of Ruin
- Equation Dragon
- The Gambler
- The Zero King

THE ASSISTANT

Vantor, the Remainder Warlock

"I was the part of an equation that got left over. It gives one perspective."

Read the Ward

free

Properties, never answers. Odd or even. Prime. Below zero. Not a whole number. Larger than any tile you hold.

Give me the first move

2 insight

The opening pair only. "Begin with 7 × 3. I will not tell you the rest."

Work it out for me

5 insight

The full expression and every step of its resolution — then you still have to cast it yourself.

He also reacts unprompted after every cast:

"You landed 51 against 105 — 54 too low. You need more from the multiplication, or one more tile."

A DELIBERATE CHOICE

The tutor is a solver, not a language model

What Vantor is

- Exhaustive search over every expression buildable from the tiles actually in your hand.
- Evaluated by the same code that judges your spell, so his advice and the game can never disagree.
- Cannot propose a move you are unable to make.
- Runs offline, instantly, at zero cost per hint.

Why not an LLM here

- A language model doing novel mental arithmetic is occasionally, confidently wrong.
- A confidently wrong maths tutor is the worst failure mode this product has.
- It would also need a network connection mid-battle.
- Better use for one: answering “why does × go first?” — explanation, not calculation.

REWARDS

Thirteen legendary artifacts

Nine are held by region bosses and drop guaranteed. Four lesser relics appear on ordinary foes about one fight in five.

⌗ **Abacus**
+1 tile every turn

− **Broken Minus**
+25% subtracting

× **Factor Stone**
+25% multiplying

÷ **Divider's Compass**
+30% dividing

½ **Phantom Half**
Grazes always 80%

φ **Golden Ratio Ring**
+20% on 3+ tiles

x **The Unknown X**
An extra reroll

⚄ **Loaded Die**
1 spell in 4 doubles

0 **Zero King's Crown**
Exact match heals 8%

||| **Tally Bracelet**
+30 max health

✎ **Warlock's Chalk**
Free hints, extra Study

◉ **Prime Lens**
Vantor reads every Ward

⚖️ **Counterweight**
First spell +60%

KINGDOM BUILDING

Seven buildings, all of them combat systems

Nothing here is decoration. Every upgrade changes what happens inside a battle.

Academy of Numbers

+20% experience per level

Geometry Workshop

+1 tile in hand each turn

Training Arena

+25 maximum health

Royal Library

An extra Study each battle

Wizard Tower of Algebra

Unlocks brackets — and precedence

Fraction Farm

+12 gold per victory

Math Laboratory

One hand reroll per battle

Upgrades cost gold and insight — and insight is earned by getting Wards exactly right, so the kingdom grows out of accuracy rather than time spent.

ECONOMY

Three currencies, three different behaviours rewarded

XP

TIME SPENT

Levels raise health and spell power. Awarded for finishing a fight, win or lose.

Gold

VICTORIES

Buys buildings and a night's rest. Scales with how deep into a region you fought.

Insight

BEING EXACTLY RIGHT

Upgrades buildings and pays the warlock. Earned per perfect resonance, so precision is the only source.

No purchase of any kind buys any of the three. Everything on this slide is earned by playing.

CURRICULUM

What the game actually asks a player to do

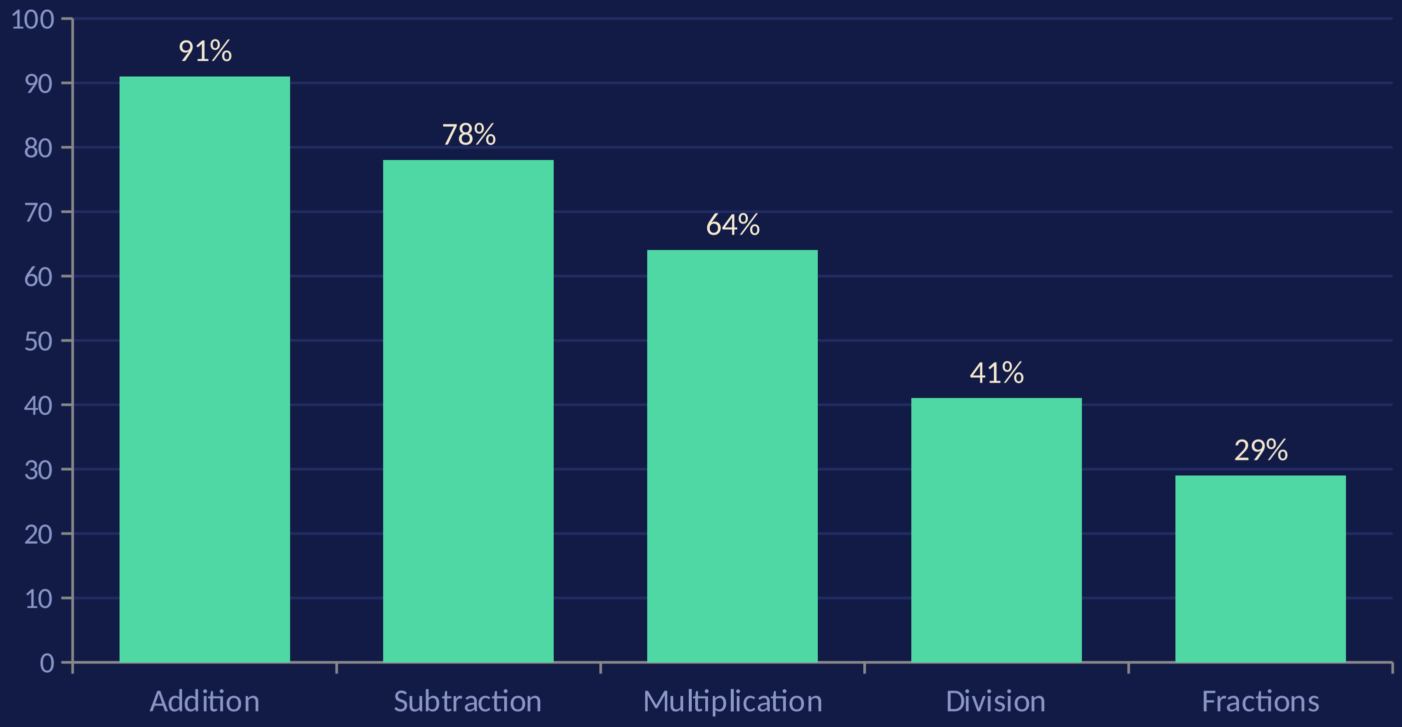
Not topics the game mentions — operations it requires before it will let you win.

Mental addition and subtraction	Every region, every turn
Times tables to 14 ×	Multiplication Mountains onward
Division and divisibility	Division Desert, Fraction Kingdom
Negative numbers	Subtraction Swamp, Final Realm
Fractions as exact quantities	Fraction Kingdom Wards arrive in halves
Order of operations	Every cast; brackets via the Tower
Primality and factors	Prime Berserker, Prime Lens, Vantor's readings
Estimation and error size	The graze band; Decimal Duelist
Area, perimeter, angle	Geometry City word Wards
Solving for an unknown	Algebra Fortress word Wards
Percentage and ratio	Probability Islands word Wards

MEASUREMENT

The Hero screen is a learning dashboard in disguise

Exact-match rate per region, updated every cast. Example profile below.



What a teacher sees

- This child is fluent in addition and shaky on division — visible in ninety seconds without a test.
- The game reads the same signal and softens division Wards automatically.
- In-game framing stays supportive: a low bar is where the next lesson is, not a failure.

HONESTY ABOUT SCOPE

What is built, and what is still a document

▶️ Playable now

- Ten classes, nine regions, twenty-seven fights.
- Full Ward and casting system with exact rational arithmetic.
- Vantor, thirteen artifacts, seven buildings, adaptive difficulty.
- Verified: 3,240 of 3,240 generated Wards provably solvable.

◌ Designed, not built

- Multiplayer, tournaments and leaderboards.
- Open-world traversal between regions rather than a menu map.
- Story, characters and voiced mentors.
- Art, animation and audio — currently a typographic placeholder.

STILL OPEN

Three questions this design has not answered

1

Is the graze band a mercy or a loophole?

The Decimal Duelist rewards estimation, which is a real and underrated skill — but it also softens the exact-match rule the whole design rests on. Good class, or a difficulty setting wearing a costume?

2

Is the hint economy regressive?

Hints cost insight. Insight is earned by getting Wards right. The players who most need help can least afford it. Motivating pressure, or a tax on struggling children?

3

Does the fun survive the maths being removed?

The test this design set itself. If a playtester enjoys the Ward puzzle but learns nothing measurable, the thesis is wrong and the whole structure needs rebuilding.

A design that cannot name its own weak points has not been examined yet.